

Merging Datasets Guided Lab

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STATES - USA Bald Eagle Screaming Noise

```
States1 <- read.csv("States1.csv")
States2 <- read.csv("States2.csv")
States <- States2 %>% left_join(States1 %>% dplyr::select(State, Union))
```

```
## Joining with `by = join_by(State)`
```

```
States<-States%>%mutate(Population = as.numeric(gsub(",", "", Population)))#Removing commas so it is numeric
head(States)
```

```
##      State MinimumWage Population BiggestCityPop NumberofCities GovernorParty
## 1  Alabama           NA    4921532      1,090,435             5      Republican
## 2  Alaska      10.34     731158      291,826             1      Republican
## 3  Arizona      12.15    7421401    1,633,017             5      Republican
## 4  Arkansas      11.00    3030522     198,606             0      Republican
## 5 California      13.00   39368078    3,990,456            15      Democrat
## 6  Colorado      12.32    5807719     705,576             3      Democrat
##      Union
## 1  1819
## 2  1959
## 3  1912
## 4  1836
## 5  1850
## 6  1876
```

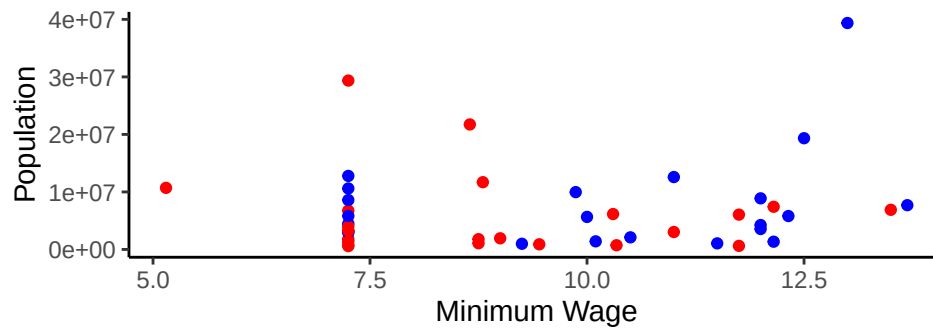
```
#Minimum Wage vs Population Faceted by Governor Population
```

```
gf_point(Population ~ MinimumWage , data = States, color = ~(GovernorParty=="Democrat")) %>%gf_refine(s
```

```
## Warning: Removed 5 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

No Relationship Between Population and Wage

Scatterplot of Minimum Wage vs Population and Party



Governor Party ● Republican ● Democrat

States 1 & 2(csv)

Animal

```
library(dplyr)
library(stringr)
library(ggplot2)
library(tidyverse)
library(scales)
library(mosaic)
Animals4 <- read.csv("Animals4.csv")
Animals5 <- read.csv("Animals5.csv")
head(Animals4)
```

```
##      Animal Lifespan.years. GestationPeriod.days. TypeofAnimal      Diet
## 1      Mole              4              40      mammal    carnivore
## 2 Capybara              4              140      mammal    herbivore
## 3 Hedgehog              5              40      mammal    insectivore
## 4  Baboon             30              180      mammal    omnivore
## 5   Horse             28              340      mammal    herbivore
## 6   Camel             40              410      mammal    carnivore
```

```
##      Length.in.
## 1              6
## 2             51
## 3             10
## 4             20
## 5             96
## 6            120
```

```
head(Animals5)
```

```
##      Animal Lifespan Gestation AvgLength NumberBabies   Class      Diet
## 1   Turtle      80       60      80.4           NA Reptilia Carnivore
## 2     Yak       20      270     126.0            1 Mammalia Herbivore
## 3 Salamander    20      913       6.4           NA Amphibia Carnivore
## 4 BlackHowler   20      180      31.2            1 Mammalia Herbivore
## 5   Vicuna      24      335      60.0            1 Mammalia Herbivore
```

```
## 6      Cabybara      10      150      48.0      5 Mammalia Herbivore
##      NumberEggs Eggs
## 1         110  Yes
## 2          NA   No
## 3         450  Yes
## 4          NA   No
## 5          NA   No
## 6          NA   No
```

```
colnames(Animals4)[c(2, 3, 6)] <- c("Lifespan", "Gestation", "Length")
Animals4 <- Animals4 %>% dplyr::select(Animal, Lifespan, Gestation, Diet, Length)
Animals5 <- Animals5 %>% dplyr::select(Animal, Lifespan, Gestation, Length = AvgLength, Diet)
Animals5 <- Animals5 %>% mutate(Diet = str_to_lower(Diet), Animal = str_replace_all(string = Animal, pat
Unique_Animals5 <- Animals5 %>% anti_join(Animals4, by = "Animal")
AllAnimals <- Animals4 %>% full_join(Unique_Animals5)
```

```
## Joining with `by = join_by(Animal, Lifespan, Gestation, Diet, Length)`
```

```
head(AllAnimals)
```

```
##      Animal Lifespan Gestation      Diet Length
## 1      Mole         4         40  carnivore      6
## 2 Cabybara         4        140  herbivore     51
## 3 Hedgehog         5         40 insectivore     10
## 4  Baboon        30        180   omnivore     20
## 5   Horse        28        340  herbivore     96
## 6   Camel        40        410  carnivore    120
```

```
nrow(AllAnimals)
```

```
## [1] 138
```

```
duplicate_animals <- AllAnimals %>%
  count(Animal) %>%
  filter(n > 1)
print(duplicate_animals)
```

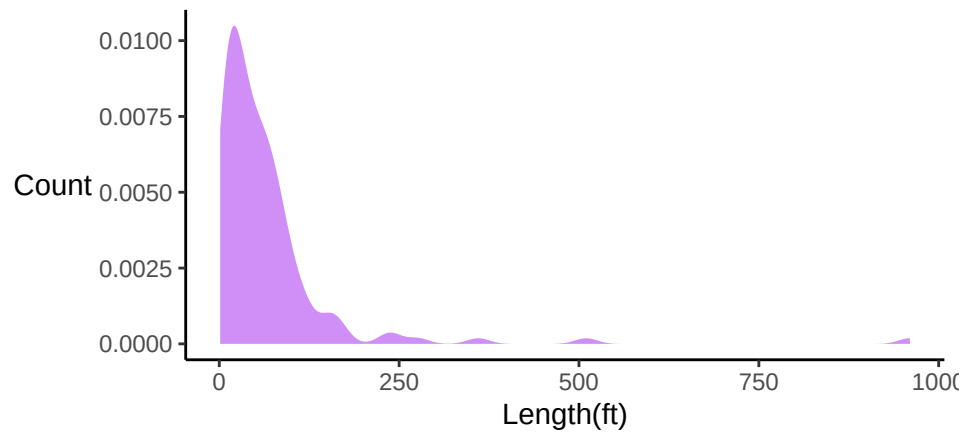
```
##      Animal n
## 1  Bison  2
```

```
#Question: How is the length of animals distributed?
```

```
gf_density(~Length, data = AllAnimals, fill = "purple", alpha = 0.5)%>%gf_labs(x = "Length(ft)", y = "C
```

The Majority of Animals are Smaller than 200

Distribution of Animal Diets



Animals 4 & 5(csv)